



Super Green Farm — Crop Production Manual

 This manual outlines standard operating procedures (SOPs) for planning, establishing, managing, harvesting, and storing crops at **Super Green Farm**.

1) Purpose and scope

- Provide consistent, repeatable practices for crop production.
- Improve yields and quality while protecting soil, water, and worker safety.
- Applies to field crops and vegetables (open field and simple irrigation systems).

2) Roles and responsibilities

- **Farm manager:** seasonal planning, input procurement, supervision, records.
- **Field supervisor:** daily task assignment, scouting, irrigation and pest control checks.
- **Operators/labor:** follow SOPs, use PPE, report issues immediately.

3) Pre-season planning

3.1 Crop selection

- Choose crops based on market demand, climate, water availability, and farm capacity.
- Prefer a mix of:

- Short-cycle crops (cash flow)
- Medium/long-cycle crops (stability)
- Legumes (soil fertility)

3.2 Cropping calendar

- Build a calendar for:
 - Land prep → planting → fertilizer topdressing → pest control → harvest
 - Expected rainfall/temperature patterns
 - Labor peaks and equipment needs

3.3 Field mapping and rotation plan

- Divide the farm into blocks/plots.
- Rotation guidelines:
 - Do not plant the same crop family in the same plot in consecutive seasons.
 - Rotate heavy feeders (e.g., maize, cabbage) with legumes (e.g., beans, groundnuts).

4) Site assessment and soil management

4.1 Soil testing

- Test soil at least once per year (or per season for intensive vegetables):
 - pH
 - Organic matter
 - N-P-K
 - Salinity (if irrigated)
- Keep results in a farm records file.

4.2 Soil improvement

- Incorporate organic matter:

- Well-composted manure
- Crop residues (where safe)
- Compost
- Protect soil structure:
 - Avoid working wet soil
 - Reduce unnecessary tillage

4.3 Erosion control

- Use contour planting, grass strips, mulching, or cover crops on sloped areas.

5) Land preparation

5.1 Clearing and initial prep

- Remove weeds and residues; compost what is appropriate.
- Mark plot boundaries and irrigation lines.

5.2 Tillage options (choose based on soil and crop)

- **Conventional tillage:** plough + harrow for fine seedbed (good for small seeds).
- **Reduced tillage:** minimum disturbance to conserve moisture and soil life.

5.3 Bed and ridge formation (vegetables)

- Raised beds improve drainage and irrigation efficiency.
- Standardize bed width for easy management (based on tools and spacing).

6) Seed and planting material management

6.1 Seed quality

- Use certified seed when possible.
- Check:

- Germination rate
- Seed health (no mold, insect damage)
- Variety suitability

6.2 Seed storage

- Store seed in a cool, dry, dark place.
- Label by crop, variety, lot, and date.

6.3 Seed treatment (when applicable)

- Follow product label instructions.
- Wear PPE and avoid contamination of water sources.

7) Planting operations

7.1 Timing

- Plant according to the cropping calendar and soil moisture conditions.

7.2 Spacing and population

- Use recommended spacing by crop/variety.
- Keep a reference table for key crops used at Super Green Farm.

7.3 Planting depth

- General guideline:
 - Small seeds: shallow
 - Large seeds: deeper
- Ensure firm soil-to-seed contact.

7.4 Transplanting (vegetables)

- Harden seedlings 5–7 days before transplanting.
- Transplant in late afternoon or on cloudy days to reduce stress.

- Water immediately after transplanting.

8) Irrigation and water management

8.1 Water source and quality

- Confirm water availability for the season.
- Check water quality if salinity is a risk.

8.2 Irrigation methods

- **Drip:** efficient, ideal for vegetables and fertigation.
- **Sprinkler:** good for many crops but higher evaporation.
- **Furrow/flood:** low cost but less efficient; manage to avoid waterlogging.

8.3 Scheduling

- Irrigate based on:
 - Crop stage (seedling/flowering/fruiting)
 - Soil type (sand vs clay)
 - Weather conditions
- Avoid:
 - Overwatering (disease, nutrient loss)
 - Water stress during flowering and grain/fruit filling

8.4 Drainage

- Ensure excess water can exit plots to prevent root diseases.

9) Nutrient management (fertility)

9.1 Principles

- Apply nutrients based on soil test results and crop needs.
- Split nitrogen applications to reduce losses.

9.2 Organic fertilizers

- Compost/manure application:
 - Apply and incorporate before planting
 - Use only well-composted manure to reduce pathogens and weed seeds

9.3 Inorganic fertilizers

- Follow local recommendations and product labels.
- Typical approach:
 - Basal application at planting (often P and some N)
 - Topdressing N at key growth stages

9.4 Micronutrients

- Correct deficiencies (e.g., zinc, boron) if identified in tests or visible symptoms.

10) Weed management

10.1 Prevention

- Use clean seed.
- Maintain field borders.
- Rotate crops.

10.2 Control methods

- **Mechanical:** hand weeding, hoeing, cultivation.
- **Mulching:** reduces weeds and conserves moisture.
- **Chemical:** herbicides only when trained and following labels.

10.3 Critical periods

- Keep fields weed-free during early crop establishment.

11) Integrated Pest and Disease Management (IPM)

11.1 Scouting

- Scout fields at least 2 times per week (more in high-risk periods).
- Record:
 - Pest/disease seen
 - Location/plot
 - Severity level
 - Action taken

11.2 Prevention

- Use resistant varieties when available.
- Remove diseased plants when practical.
- Practice rotation and field hygiene.
- Manage irrigation to reduce humidity-related diseases.

11.3 Threshold-based action

- Treat only when pest levels justify it.

11.4 Safe pesticide use (if used)

- Use PPE (gloves, mask/respirator as required, boots, long sleeves).
- Respect pre-harvest intervals (PHI) and re-entry intervals (REI).
- Calibrate sprayers.
- Keep pesticides locked and labeled.

12) Crop monitoring and field operations

12.1 Routine checks

- Germination/stand count

- Plant vigor and color
- Pest and disease pressure
- Moisture status
- Nutrient deficiency symptoms

12.2 Gap filling and thinning

- Replace missing plants early.
- Thin where plant density is too high.

12.3 Staking, pruning, and trellising (where relevant)

- Apply for tomatoes, climbing beans, cucumbers, etc.

13) Harvest management

13.1 Harvest timing

- Harvest at correct maturity for the market (fresh vs processing).
- Avoid harvesting when produce is wet to reduce rot.

13.2 Harvest hygiene and handling

- Use clean crates and tools.
- Keep produce shaded.
- Minimize bruising.

13.3 Field grading (basic)

- Separate:
 - Marketable
 - Damaged/diseased
 - Undersized/overmature

14) Post-harvest handling and storage

14.1 Cooling (where possible)

- Rapidly remove field heat (shade, ventilation, evaporative cooling, cold room if available).

14.2 Cleaning and packaging

- Wash only when necessary and with clean water.
- Use appropriate packaging to prevent damage.

14.3 Storage

- Store by commodity requirements:
 - Temperature
 - Humidity
 - Ventilation
- Keep storage areas clean and pest-free.

15) Food safety, worker health, and environmental safety

- Train workers on hygiene and safe handling.
- Provide clean drinking water and handwashing.
- Use PPE for chemicals and dusty operations.
- Dispose of chemical containers safely (never reuse for food/water).
- Prevent contamination of rivers, wells, and irrigation canals.

16) Records and documentation (recommended templates)

Maintain simple records for:

- Land preparation and planting dates
- Seed lots and sources

- Input use (fertilizers, pesticides)
- Irrigation dates and amounts (or hours)
- Scouting and treatment logs
- Harvest quantities and grades
- Sales and buyers

17) Emergency and problem response

- **Drought/water shortage:** prioritize high-value crops; adjust planting density; increase mulching.
- **Flooding/waterlogging:** improve drainage immediately; replant if stand loss is severe.
- **Pest outbreak:** intensify scouting; isolate affected plots; apply IPM actions quickly.
- **Input shortage:** prioritize critical stages (e.g., topdress at flowering); seek alternatives (compost).

18) Season review and continuous improvement

- After harvest, review:
 - Yield by plot
 - Costs and profit
 - Pest/disease issues
 - What worked / what didn't
- Update next season's calendar and SOPs.

✓ **Next step:** list the main crops grown at Super Green Farm (e.g., maize, sorghum, tomatoes, onions, okra, beans) and add crop-specific pages with recommended spacing, fertilizer schedule, and common pests.